Craig Lionel Roberts

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Education

Northeastern University — Khoury College of Computer Sciences Sep 2024 – Dec 2026
MS in Artificial Intelligence; GPA: 3.81/4.0 Boston, MA

- Coursework: Machine Learning, Natural Language Processing, Algorithms, Linear Algebra and Probability for Data Science, Foundations of AI, Deep Learning

VIT Bhopal University Jul 2020 – May 2024
B.Tech in CS with AI and ML; GPA: 3.52/4.0 Bhopal, India

- Coursework: Data Structures, Database Management Systems, Discrete Mathematics, Foundations of Machine Learning

Technical Skills

LLM and Agent Systems: Multi-Agent Orchestration, LLM Supervisor & Router Architectures, Tool Use & Function Calling, Prompt Engineering, Guardrails, Structured Output Extraction
RAG and Retrieval: Retrieval-Augmented Generation, Hybrid Lexical & Vector Search, Azure AI Search, Vector Databases (Pinecone, ChromaDB), Reranking, Embedding Models (EmbeddingGemma, Azure OpenAI Embeddings)
ML and Modeling: PyTorch, TensorFlow, scikit-learn, HuggingFace (Transformers, PEFT/LoRA), Sentence-BERT, Deep Learning
MLOps and Infrastructure: Databricks (Model Serving, Apps, Unity Catalog), FastAPI, asyncio, Azure DevOps CI/CD, REST APIs
Languages and Frameworks: Python, SQL, JavaScript, C++, Java, Git

Experience

Staples, Inc. — ML Core Team Jan 2026 – Aug 2026
Applied AI Engineer Co-op Framingham, MA

- Built and maintained **Sparky**, a production **RAG** knowledge assistant grounding responses in 8,000+ internal articles via **Azure AI Search**, then fixed a context-window bug that cut token usage per turn from 188K to 1.7K.
- Re-embedded the knowledge base with **EmbeddingGemma-300M** to widen retrieval coverage and built a **reranker** as part of a two-person team after tracing weak answers back to retrieval quality rather than the underlying model.
- Architected a 5-**agent orchestration** platform on **Databricks Model Serving** where an **LLM** supervisor classifies intent and routes cases to specialized agents concurrently via **asyncio**, supporting 14 business action types, working alongside one senior engineer.
- Debugged production issues across the platform: root-caused a dev/QA library mismatch in `deploy_model.ipynb`, fixed null trace attribution affecting 42% of **MLflow** traces, and resolved a deploy cap via an asynchronous polling fix.
- Onboarded the incoming Fall co-op through structured knowledge-transfer sessions on the `general_parser` and **Sparky** codebases, and partnered with stakeholders across sales, operations, and customer service to translate requirements into production **AI** systems.

DeepRoot Minds Technologies Jun 2023 – Dec 2023
Machine Learning Intern Remote

- Fine-tuned pretrained **CNN** vision models via transfer learning on 3,000+ dermatological images, improving skin cancer classification performance by 12% through targeted layer unfreezing and optimized preprocessing.
- Constructed a **GAN**-driven augmentation pipeline enhancing dataset diversity, increasing generalization capability by 10% across evaluation sets, with **MLflow** experiment tracking.

Projects

AEO Content Engine | *Python, FastAPI, ChromaDB, Streamlit, MLflow* Jul 2026 – Aug 2026

- Engineered a 4-step chained **LLM** generation pipeline independently, grounded in a **ChromaDB** retrieval corpus with inline citations and served through an 8-endpoint **FastAPI** service with a **Streamlit** monitoring dashboard.
- Wrote an **LLM**-as-judge evaluation module scoring output on factuality, coverage, tone, and originality, catching regressions and routing anything below threshold to a review queue before publication.
- Clustered 360 seed questions into 10 topics using sentence-transformers and **BERTopic**, tracking coherence scores through **MLflow** to validate topic quality before generation.

Candidate Recommender System | *Sentence-BERT, PyTorch, NLP* | GitHub Aug 2025 – Sep 2025

- Trained a **Sentence-BERT** ranking model on candidate-job pairs with contrastive learning, reaching 89% precision@5 across 100+ benchmark queries with sub-3-second scoring.
- Diagnosed failure modes across query distributions and refined training-data curation and loss formulation to improve generalization on unseen candidate-job pairs.
- Wrote evaluation scripts to benchmark ranking performance against held-out queries, catching regressions before merging changes into the scoring pipeline.

NomadAI | *RAG, Pinecone, Streamlit* | GitHub Jul 2025 – Aug 2025

- Assembled an end-to-end **RAG** pipeline indexing 2,400+ Reddit discussions into **Pinecone** via dense embedding retrieval, tuning chunk size and top-k to balance recall against token budget.
- Integrated Reddit and Google Places **APIs** into a scalable ingestion pipeline and shipped a full **Streamlit** application on top of it, handling location-aware queries end to end.
- Resolved retrieval-quality issues in the location-aware lookup layer by iterating on chunking and ranking parameters, raising relevance across the recommendation set.